Exercise for the course on Creating Electronic Books: Multiloop techniques in Particle Physics

Course for the Faculties of Sciences and Chemical Sciences

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University of Salamanca

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CONTENTS

Welcome to my assignment for the course **Creating Electronic Books with Code and Artificial Intelligence Assistants**.

What is this?

This is my assignment for the course based on the template designed for the teaching staff of the **Faculties of Sciences and Chemical Sciences at USAL**. The main purpose of this page is to demonstrate basic handling of the content learned during the course.

PDF version

You can also download the printable version of the book:

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Part I

Example

WHAT IBP MEANS FOR FEYNMAN INTEGRALS

In the context of Feynman integrals, integration-by-parts (IBP) identities are linear relations obtained from integrals of total derivatives in loop-momentum space within dimensional regularisation. They relate integrals with different propagator powers and allow reduction of any integral in a family to a finite set of master integrals. Once master integrals are identified, one typically constructs differential equations for them and attempts to transform them to an ε-form to solve them in terms of iterated integrals.

1.1 Definition — Feynman integral:

A Feynman integral associated to a graph G with l loops and n propagators is commonly written as

$$I(\nu_1, \dots, \nu_n) = \int \prod_{j=1}^{l} \frac{d^D k_j}{i\pi^{D/2}} \, \frac{1}{D_1^{\nu_1} \cdots D_n^{\nu_n}} \,,$$

where the D_i denote inverse propagators and the integers ν_i their powers.

1.2 Definition — IBP (integration by parts) for Feynman integrals:

Within dimensional regularisation the integral of a total derivative vanishes (no boundary terms). For any loop momentum k_i and any vector q constructed from external and loop momenta:

$$0 = \int \prod_{j=1}^{l} d^D k_j \, \frac{\partial}{\partial k_i^{\mu}} \Big\{ q^{\mu} \, \frac{1}{D_1^{\nu_1} \cdots D_n^{\nu_n}} \Big\} \,.$$

Expanding this equation yields linear relations among Feynman integrals with shifted indices ν_j; these are the IBP identities. By generating sufficiently many identities and applying an ordering criterion (Laporta algorithm) one reduces the family to a finite basis of master integrals.

1.3 Key consequences and workflow:

- Reduction: IBP identities allow reduction of generic integrals to master integrals by algebraic elimination.
- Master integrals: only the master integrals need to be computed explicitly; they form a basis that depends on the chosen ordering criterion.
- Tools: public programs such as FIRE, Reduze and Kira perform IBP reductions, using finite-field methods and sparse linear algebra to improve performance.

Example (sketch):

For the one-loop two-point function with equal masses, IBP relations allow reduction of integrals with indices (ν_1, ν_2) to the masters $I_{1,1}$ and $I_{1,0}$ (bubble and tadpole). See the referenced chapter for the full derivation and diagrams; the bubble and tadpole topologies are shown in Fig. 2.1 and Fig. 2.2.

INTEGRATION BY PARTS (IBP) AND KIRA / FIRE TOOLS

2.1 What IBP means for Feynman integrals

In the context of Feynman integrals, integration-by-parts (IBP) identities are linear relations obtained from integrals of total derivatives in loop-momentum space within dimensional regularisation. They relate integrals with different propagator powers and allow reduction of any integral in a family to a finite set of master integrals. Once master integrals are identified, one typically constructs differential equations for them and attempts to transform them to an ε-form to solve them in terms of iterated integrals.

2.1.1 Definition — Feynman integral:

A Feynman integral associated to a graph G with l loops and n propagators is commonly written as

$$I(\nu_1, \dots, \nu_n) = \int \prod_{j=1}^{l} \frac{d^D k_j}{i\pi^{D/2}} \, \frac{1}{D_1^{\nu_1} \cdots D_n^{\nu_n}} \, ,$$

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Expanding this equation yields linear relations among Feynman integrals with shifted indices ν_j; these are the IBP identities. By generating sufficiently many identities and applying an ordering criterion (Laporta algorithm) one reduces the family to a finite basis of master integrals.

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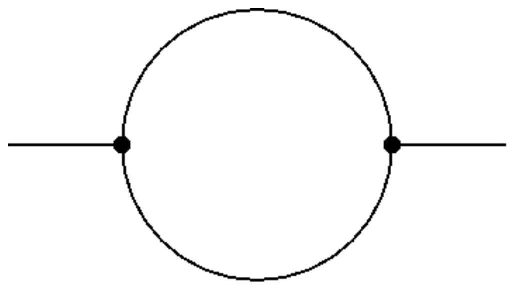

Figure2.1: Bubble (one-loop two-point) topology.

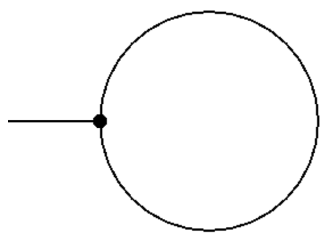

Figure2.2: Tadpole (one-loop one-point) topology.

2.1.4 Remarks:

- Different ordering choices (dot-basis, ISP-basis) lead to different practical master sets.
- IBP reduction is algebraic over rational functions in kinematic variables and the dimension D; simplification of rational functions is typically the performance bottleneck.

Reference (APA): Weinzierl, S. (2022). Feynman Integrals. arXiv:2201.03593. https://arxiv.org/abs/2201.03593

2.2 Using FIRE6 for IBP reduction

FIRE6 is a public program to reduce Feynman integrals to master integrals; it implements modular-arithmetic reductions and a C++ backend designed for large problems. Typical workflow: prepare a Mathematica "start" file describing the family (momenta, propagators, replacements), optionally generate LiteRed rules, save the start file and run the C++ FIRE6 reduction with a small config file.

Minimal example (start file in Mathematica):

```
Get["FIRE6.m"];
Internal = {k1, k2};
External = {p1, p2, p3};
Propagators = {-k1[2], -(k1 + p1 + p2)[2], -k2[2], ...(etc.)};
Replacements = {p1[2] -> 0, p2[2] -> 0, p1 p2 -> s/2};
PrepareIBP[];
Prepare[];
SaveStart["doublebox"]; Quit[];
```

Example C++ config file (`doublebox.conf`):

```
#threads            4
#fthreads           4
#variables          d,s,t
#start
#folder             examples/
#problem            1 doublebox.start
#integrals          doublebox.m
#output             ../tests/outputs/doublebox.tables
```

Run with:

```
bin/FIRE6 -c examples/doublebox
```

The file `doublebox.tables` contains the reduced expressions (integrals expressed in terms of masters) which can be loaded into Mathematica with `LoadTables`.

See the FIRE6 manual (`recursos/FIRE6.pdf`) for installation and advanced options.

Citation: Smirnov, A.V. & Chukharev, F.S. (2020). FIRE6: Feynman Integral REduction with modular arithmetic. Computer Physics Communications, 247, 106877. DOI: 10.1016/j.cpc.2019.106877

Part II

Information

ABOUT

You can cite this book as:

> Álvaro Lozano Murciego and André Sales Mendes (2025). *Creating Electronic Books with Code and Artificial Intelligence Assistants*. https://elloza.com/teachbook_usal_template/ . Source files at https://github.com/elloza/teachbook_usal_template . CC BY 4.0.

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> `<Page title>`. In Álvaro Lozano Murciego and André Sales Mendes (2025). *Creating Electronic Books with Code and Artificial Intelligence Assistants*. https://elloza.com/teachbook_usal_template/ . Source files at https://github.com/elloza/teachbook_usal_template . CC BY 4.0.

Because this book is expected to evolve over time, it is recommended to cite the source repository as well whenever the exact context of the material matters.

3.1 How this book is made

This website is written in Markdown files and Jupyter notebooks, and converted to HTML using tools from TeachBooks and Jupyter Book.

The source files are stored in a public GitHub repository:

- https://github.com/elloza/teachbook_usal_template

The published version of the book can be viewed at:

- https://elloza.com/teachbook_usal_template/

3.2 About the authors

3.2.1 Authors

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3.2.2 Institution

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CREDITS AND LICENSES

4.1 Project credits

This book and the course template are built with TeachBooks, on top of the Jupyter Book and MyST Markdown ecosystem.

TeachBooks provides documentation, components and working patterns that make it easier to create open, interactive and reproducible teaching books.

4.2 Acknowledgements

We thank the TeachBooks project and Delft University of Technology (TU Delft) for their work on open tools for educational books. The credits page in the TeachBooks Manual is used as a reference for explicitly acknowledging the tools, communities and institutional support on which this material builds.

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- TeachBooks
- Jupyter Book
- MyST Markdown

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5.1 Text Citation

Faculties of Sciences and Chemical Sciences (2026). *Creating Electronic Books with Code and Artificial Intelligence Assistants*. University of Salamanca. Available at: https://github.com/elloza/teachbook_usal_template

5.2 BibTeX

```
@book{creating_electronic_books_ai_2026,
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  title = {Creating Electronic Books with Code and Artificial Intelligence Assistants}
↪,
  year = {2026},
  publisher = {University of Salamanca},
  url = {https://github.com/elloza/teachbook_usal_template}
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```

5.3 DOI

(Pending assignment via Zenodo)

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